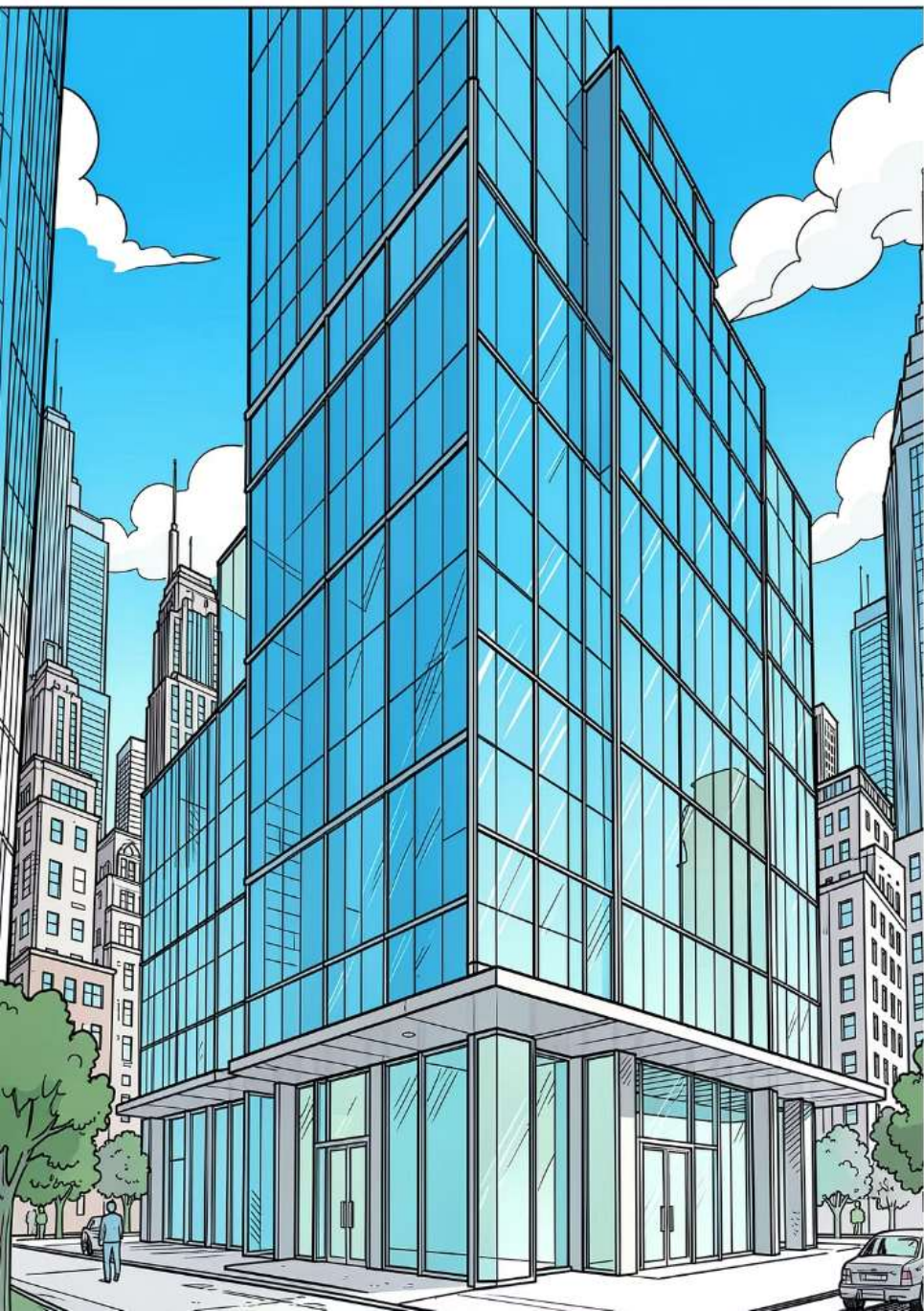




DATA ANALYTICS PROJECT

MAGADA BANK Loan Portfolio Analysis Dashboard

Created By- Subhayan Mitra & Debtanu De

An interactive Excel-based analytics solution analyzing 1,000 loan records to drive data-driven decisions in credit risk, portfolio performance, and customer assessment.

The Challenge: Managing Loan Portfolio Risk



Business Problem

Magada Bank lacked a centralized view of its loan portfolio, making it difficult to monitor defaults, assess creditworthiness, and optimize lending strategies in real time.

Project Objectives

- Analyze loan distribution, interest rates, and repayment status
- Assess customer creditworthiness using CIBIL scores
- Build an interactive, filterable dashboard in Excel
- Enable data-driven risk and portfolio decisions

Dataset Overview & Data Cleaning

1000

Loan Records

Customer loan entries analyzed

₹17.48Cr

Portfolio Value

Total loan disbursement value

₹17.48L

Avg Loan Size

Mean loan amount per customer

725

Avg CIBIL Score

Customer creditworthiness indicator

10%

Avg Interest Rate

Mean rate across all loan types

Raw Data Issues Identified

- Duplicate records and null values
- Inconsistent date formats and text casing
- Misclassified loan types and status labels

Power Query Transformations

- Removed duplicates and trimmed whitespace
- Standardized columns via Change Type & Replace Values
- Merged customer tables using Merge Queries
- Created custom columns for risk categorization

Excel Features Powering the Dashboard



Pivot Tables & Charts

Aggregated loan data by type, status, and region for dynamic summary views.



XLOOKUP & VLOOKUP

Mapped customer IDs to credit profiles and enriched records with external data.



Interactive Slicers

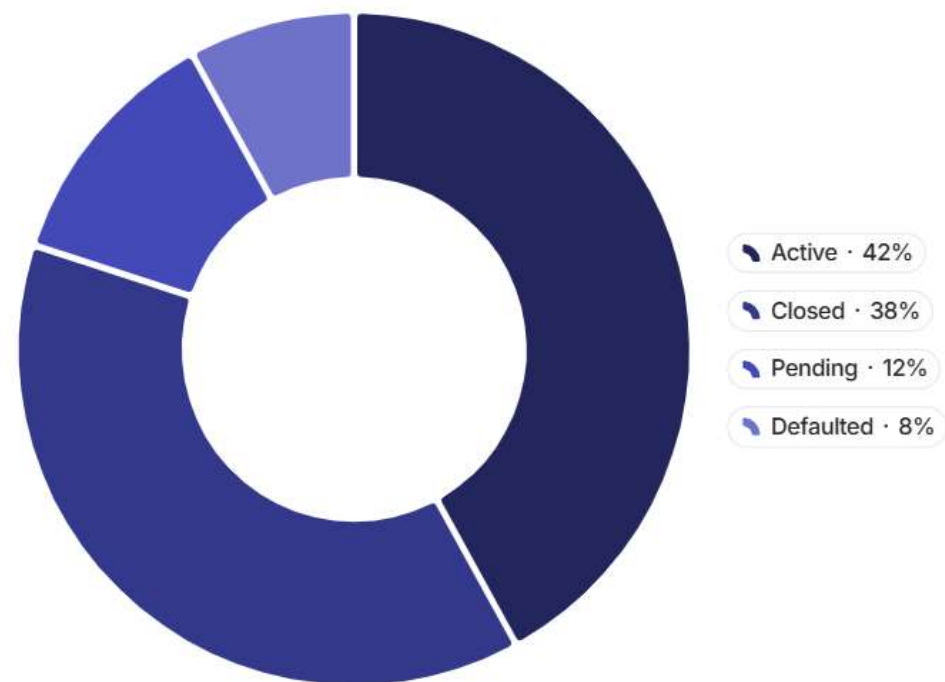
Filter dashboard visuals by loan type, status, and CIBIL range instantly.



Power Query

Automated ETL pipeline — import, clean, transform, and load with one refresh.

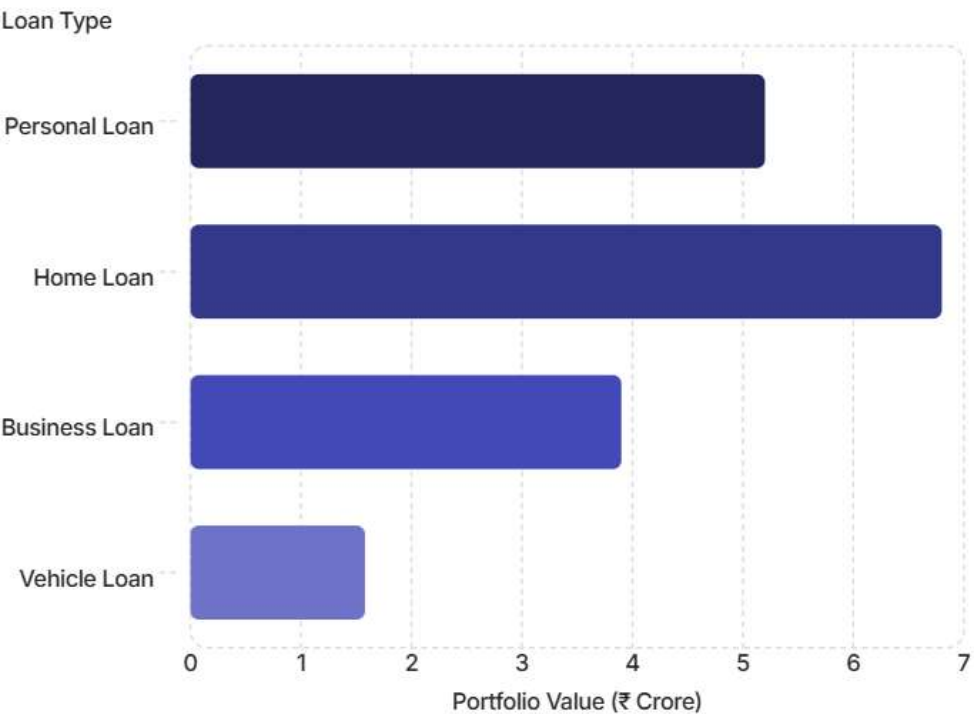
Loan Status & Portfolio Composition



Loan Status Breakdown

80% of loans are performing (Active + Closed), while 8% are defaulted — a key risk indicator requiring attention.

Portfolio by Loan Type



What the Data Reveals

Interest Rate vs. Loan Amount

Higher loan amounts correlate with lower interest rates — premium customers receive preferential pricing, reflecting stronger credit profiles.

CIBIL Score Distribution

Customers with CIBIL scores above 750 show a **94% repayment rate**, while scores below 650 carry a **3x higher default risk**.

Home Loans Dominate Portfolio

Home loans represent **39% of total portfolio value** (₹6.8 Cr), making them the highest-value segment with the lowest default rate.

Default Concentration

Personal loans account for **62% of all defaults**, signaling a need for stricter credit assessment in this segment.

Recommendations & Future Scope

Business Recommendations

- **Risk Management:** Introduce CIBIL-based tiered interest rates to reduce default exposure
- **Credit Assessment:** Flag personal loan applicants with scores below 680 for manual review
- **Portfolio Optimization:** Increase home loan disbursements — highest value, lowest risk
- **Early Warning:** Monitor "Pending" loans (12%) proactively to prevent escalation

Future Scope

- Integrate Power BI for real-time dashboard publishing
- Add macro-driven scenario analysis for stress testing
- Connect to live banking APIs for automated data refresh
- Build predictive default models using Excel's statistical tools



Conclusion: This dashboard demonstrates how Excel can serve as a powerful analytics engine for banking — enabling faster, smarter, data-driven lending decisions.